Program analysis for machine learning and comparison tables

Program analysis for ML

Several systems address the intersection of program analysis and machine learning that COGANT operates in:

code2seq and code2vec [@alon2019code2vec] represent programs as sets of AST paths between terminal nodes. These path-based representations are effective for method naming and code summarization but discard the full graph topology that graph-neural-network-based models exploit. COGANT preserves the complete program graph — including control-flow and data-flow edges — and can export it in tensor form, enabling downstream architectures that reason over richer relational structure.

Learning to Represent Programs with Graphs [@allamanis2018learning] is the closest conceptual neighbour: it constructs typed graphs fusing AST, control, and data-flow edges and feeds them to a gated graph neural network for variable-misuse and variable-naming tasks. COGANT's program graph IR generalises that construction into a pluggable and